Ministry of Science and Higher Education of the Russian Federation
Federal State Autonomous Educational Institution of Higher Education

"National Research University ITMO"

«Faculty of Control Systems and Robotics»

Field of study (specialty) 15.04.06 – Robotics and artificial intelligence

REPORT

on Research Internship

Assignment topic: Controller Design and Multi-Agent Evaluation Framework for Constrained Irrigation Optimization

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Saint Petersburg

2026

Abstract

This report presents the complete mathematical formulation, architectural design, and evaluation framework for the four irrigation controllers deployed on a 130-agent topographical agricultural field in Gilan Province, Iran. The controllers span a hierarchy of increasing sophistication: a rainfed no-irrigation baseline, a fixed-schedule heuristic reflecting traditional Gilan practices, a Model Predictive Controller (MPC) formulated as a large-scale Non-Linear Program (NLP) solved via the IPOPT interior-point method within the CasADi symbolic framework, and a Soft Actor-Critic (SAC) reinforcement learning agent trained under a Centralized Training with Decentralized Execution (CTDE) paradigm. Each controller addresses the same constrained finite-horizon optimal control problem: maximizing terminal Hashemi rice biomass over a 93-day cultivation season while strictly adhering to a seasonal water budget of 484 mm at full allocation, with deficit scenarios at 85% and 70%. The report formalizes the multi-objective cost function with five normalized penalty terms, details the Control-Oriented Model (COM) including the smooth approximation techniques required for solver convergence, specifies the NLP structure (3,120 decision variables, 2,081 constraints at prediction horizon $H_p = 8$), documents the IPOPT solver configuration and termination criteria, and presents the Markov Decision Process formulation for the SAC agent. A unified evaluation architecture ensuring identical environmental conditions across all controllers is established, together with a formal meteorological forecast uncertainty model. This chapter serves as the methodological bridge between the virtual plant described in the preceding system description and the comparative experimental results that follow.

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1 Introduction

The preceding chapter established the high-fidelity Agent-Based Model (ABM) that serves as the virtual plant for this research: a 130-agent crop-soil simulation environment on a topographically heterogeneous field in Gilan Province, Iran, with five coupled state variables per agent, cascade surface water routing governed by a directed elevation graph, and validated hydrological dynamics (Pearson $r = 0.738$ against NASA GWETROOT satellite observations in the dry-year scenario). With the physical environment rigorously defined and parameterized for Hashemi rice cultivation, the present chapter addresses the central engineering question: *how should the irrigation control policy be designed to maximize crop yield under strict water resource constraints, and how do model-based and learning-based approaches compare in this setting?*

The agricultural irrigation problem possesses several characteristics that make it particularly challenging from a control-theoretic perspective. First, the system dynamics are nonlinear and spatially coupled: surface runoff cascades through the directed elevation graph, creating complex interdependencies between the 130 agents that a controller must anticipate. Second, the system is subject to significant stochastic disturbances in the form of unpredictable precipitation events, which can range from zero rainfall over extended dry periods to concentrated storm events exceeding 20 mm/day. Third, the problem is subject to hard constraints at multiple scales: a per-agent actuator limit of 12.0 mm/day imposed by the physical irrigation infrastructure, and a global seasonal water budget that reflects regional water scarcity. Finally, the objective is inherently multi-criteria, requiring the controller to balance biomass maximization against water conservation, drought prevention, and waterlogging mitigation.

This chapter is organized as follows. Section 2 formalizes the constrained optimal control problem, defining the state space, action space, system dynamics, and the general cost structure. Section 3 presents the two open-loop baseline controllers that serve as lower-bound performance references. Section 4 details the Model Predictive Control architecture, including the multi-objective cost function, the Control-Oriented Model with its smooth approximation techniques, the NLP construction and variable structure, the IPOPT solver configuration with precise termination criteria, the prediction horizon selection, and the weight calibration methodology. Section 5 describes the Soft Actor-Critic reinforcement learning architecture, including the Markov Decision Process formulation, the Gymnasium environment wrapper, the CTDE network structure, the budget constraint handling mechanism, and the training protocol. Section 6 establishes the unified evaluation framework, including the meteorological forecast uncertainty model, the complete controller taxonomy, and the experimental grid. Section 7 concludes with a summary of the design decisions and their theoretical trade-offs.

2 Problem Formulation and System Constraints

The intelligent irrigation management of the 130-agent topographical grid is formally defined as a constrained, finite-horizon optimal control problem. Over the $K = 93$ day cultivation season for Hashemi rice, the control architecture must dynamically balance the conflicting objectives of

maximizing terminal crop yield against the cost of water consumption, while strictly adhering to the physical limitations of the irrigation hardware and the regional water allocation. To ensure a standardized comparative evaluation across both traditional heuristics and advanced learning-based algorithms, the system state, action space, constraint structure, and transition dynamics must be rigorously defined.

2.1 State Space Representation

The discrete-time system state at day k, denoted as $\mathbf{X}_k$, aggregates the physical and biological conditions of all $N = 130$ spatial agents. The state space $\mathcal{X}$ is a multi-dimensional continuous space where each agent $n \in \{1, 2, \ldots, N\}$ is described by a state vector x_k^n encompassing both hydrological and phenological variables:

$$\mathbf{X}_k = \{x_k^1, x_k^2, \ldots, x_k^N\} \in \mathcal{X} \tag{1}$$

where each x_k^n consists of five state variables:

- x_1^n: Root-zone soil moisture (mm), bounded by $[WP \cdot \theta_5, \theta_{sat} \cdot \theta_5] = [60, 200]$ mm.
- x_2^n: Accumulated thermal time (°C·days), monotonically increasing from $x_{2,\text{init}} = 210$ to maturity threshold $\theta_{18} = 1250$.
- x_3^n: Crop maturity index (dimensionless), tracking cumulative heat and drought stress penalties.
- x_4^n: Accumulated aboveground dry biomass (g/m^2), initialized at $x_{4,\text{init}} = 60$ g/m^2 reflecting the 20-day nursery period.
- x_5^n: Surface ponding depth (mm), representing un-infiltrated water remaining on the soil surface.

The full system state therefore lies in $\mathbb{R}^{5N} = \mathbb{R}^{650}$. To accurately simulate the initial flooded condition required for freshly puddled rice fields in Gilan, the system enforces a strict state initialization constraint across the entire spatial grid. At $k = 0$, the root-zone soil moisture is uniformly initialized to 100% field capacity:

$$x_1^n(0) = \theta_6 \cdot \theta_5 = 0.35 \times 400 = 140 \text{ mm} \quad \forall n \in \{1, \ldots, N\} \tag{2}$$

with $x_5^n(0) = 0$ mm (no initial ponding). The thermal time and biomass are initialized at their nursery-calibrated values $x_2^n(0) = 210$ and $x_4^n(0) = 60$ g/m^2 respectively, while $x_3^n(0) = 0$.

2.2 Action Space and Actuator Constraints

The control action at day k, denoted as $\mathbf{U}_k$, dictates the daily irrigation depth applied to each agent. The action space $\mathcal{U}$ is bounded by the physical realities of the pumping and distribution infrastructure. Let u_k^n represent the irrigation command dispatched to agent n on day k. The controller is strictly constrained by a maximum daily actuator limit, $u_{\max}$, establishing a bounded continuous action space:

$$\mathcal{U} = \left\{ \mathbf{U}_k \in \mathbb{R}^N \mid 0 \leq u_k^n \leq u_{\max} \quad \forall n \in \{1, \ldots, N\} \right\} \tag{3}$$

Based on the physical capacities of precision drip infrastructure and the hydraulic constraints of the Gilan field, the actuator cap is defined as $u_{\max} = 12.0$ mm/day. This value represents the maximum volumetric delivery rate achievable by a single irrigation zone without exceeding the soil infiltration capacity or generating excessive surface runoff. Any policy π or optimization solver must ensure $\mathbf{U}_k \in \mathcal{U}$ at every discrete time step.

2.3 Global Water Budget Constraint

A defining feature of the constrained irrigation problem, and the primary motivation for this research in the context of Iran's anthropogenic drought crisis, is the presence of a hard global resource constraint. The total cumulative irrigation volume applied across all agents over the entire growing season must not exceed a finite seasonal water budget W_{total}:

$$\sum_{k=0}^{K-1} \sum_{n=1}^{N} u_k^n \leq W_{\text{total}} \cdot N \tag{4}$$

where W_{total} is expressed in mm (equivalent depth per agent). The full irrigation requirement for Hashemi rice, derived from the 25-year climatological water budget analysis, is $W_{\text{full}} = 484$ mm. To evaluate controller robustness under increasing water scarcity, three budget tiers are defined:

Table 1: Seasonal water budget tiers for Hashemi rice.

Budget Tier	**Fraction of W_{full}**	**W_{total} (mm)**	**Scarcity Context**
Full allocation	100%	484.0	Nominal water availability
Moderate deficit	85%	411.4	Regional quota reduction
Severe deficit	70%	338.8	Drought-year rationing

The budget values are anchored to the agronomic water demand calculation $\text{ET}_c = \text{ET}_0^{\text{PM}} \cdot K_c \cdot N_{\text{days}} = 5.02 \times 1.15 \times 93 = 537.0$ mm, minus the 25-year average seasonal rainfall of 53.3 mm, yielding a full irrigation need of $I_{\text{full}} \approx 484$ mm. The 85% and 70% tiers reflect realistic policy scenarios: the moderate deficit corresponds to a 15% volumetric quota reduction that water authorities in Iranian provinces have historically imposed during low-flow years, while the severe deficit represents the extreme rationing required during consecutive drought years.

This constraint is fundamentally different in character for MPC versus RL. The MPC enforces it as a hard linear inequality constraint within the NLP formulation (Section 4.4), guaranteeing mathematical satisfaction. The RL agent, lacking native constrained-optimization mechanisms, must learn to respect the budget through reward shaping, action clipping, and early termination penalties (Section 5.4).

2.4 System Dynamics and Optimal Control Objective

The system transitions from state $\mathbf{X}_k$ to $\mathbf{X}_{k+1}$ via the nonlinear system dynamics f_{system}, which encompass both the topographical cascade routing and the biological growth models defined in the preceding chapter:

$$\mathbf{X}_{k+1} = f_{\text{system}}(\mathbf{X}_k, \mathbf{U}_k, \mathbf{W}_k) \tag{5}$$

where $\mathbf{W}_k$ represents the stochastic meteorological disturbance vector at day k, comprising rainfall $P(k)$, reference evapotranspiration $\text{ET}_0(k)$, mean/max/min temperatures $T_{\text{mean}}(k)$, $T_{\max}(k)$, $T_{\min}(k)$, and solar radiation $R_s(k)$. The overarching objective for any deployed controller is to find a policy $\pi(\mathbf{X}_k) \to \mathbf{U}_k$ that minimizes an expected cumulative cost function J over the finite horizon K:

$$\min_{\pi} \mathbb{E}_{\mathbf{W}} \left[\Phi(\mathbf{X}_K) + \sum_{k=0}^{K-1} L(\mathbf{X}_k, \mathbf{U}_k) \right] \tag{6}$$

subject to $\mathbf{U}_k \in \mathcal{U}$ and the global budget constraint (Eq. 4), where $\Phi(\mathbf{X}_K)$ represents the terminal Mayer cost (maximizing the terminal biomass x_4 at harvest), and $L(\mathbf{X}_k, \mathbf{U}_k)$ represents the intermediate Lagrange path costs penalizing cumulative water expenditure, localized ponding (x_5), drought stress, and actuator oscillation. This general optimal control formulation serves as the architectural foundation. The subsequent sections detail how this objective is uniquely approximated and solved by the baseline heuristics, the predictive optimizer (MPC), and the reinforcement learning agent (SAC).

3 Baseline and Heuristic Controllers

To rigorously evaluate the performance of the advanced closed-loop predictive (MPC) and learning-based (SAC) algorithms, two open-loop baseline controllers are implemented. These baselines represent the lower bound of crop survival and the standard agricultural heuristic currently deployed in Gilan Province, respectively. Since these are open-loop heuristics, they calculate the irrigation trajectory strictly as a function of time and total seasonal water budget, remaining entirely blind to both the daily topographical field state ($\mathbf{X}_k$) and the meteorological forecast.

3.1 The No-Irrigation Baseline (C_1)

The No-Irrigation controller (C_1) serves as the absolute lower-bound performance metric (the rainfed benchmark). It strictly enforces a zero-action policy throughout the 93-day season:

$$u_k^n = 0 \quad \forall n \in \{1, \ldots, N\}, \quad \forall k \in \{0, \ldots, K-1\} \tag{7}$$

Evaluating the field dynamics under C_1 isolates the natural hydrological capacity of the Gilan terrain, establishing a baseline terminal biomass and baseline stress duration against which the value-add of active irrigation can be quantified. This controller consumes zero water budget by definition and provides the lower bound for the yield–water trade-off frontier. Any controller that fails to significantly outperform C_1 would indicate either a flawed cost function or an ineffective optimization algorithm.

3.2 The Fixed-Schedule Heuristic (C_2)

Traditional rice cultivation in the Gilan region relies on a front-loaded irrigation strategy, providing heavy water application during the immediate post-transplanting establishment phase and linearly tapering off as the crop approaches maturity and harvest. To formalize this heuristic into a repeatable mathematical baseline, the Fixed-Schedule controller (C_2) distributes the total seasonal water budget (W_{total}) across $K_e = 19$ discrete irrigation events. The events are uniformly spaced at an interval of 5 days. To achieve the front-loaded distribution, a linear-decay weight w_j is assigned to each event $j \in \{1, 2, \ldots, K_e\}$:

$$w_j = \frac{2(K_e - j + 1)}{K_e(K_e + 1)} \tag{8}$$

By definition, $\sum_{j=1}^{K_e} w_j = 1$. The total volumetric allocation for event j is computed as $w_j \times W_{\text{total}}$. Rather than applying this volume in a single daily deluge—which would violate physical infrastructure limits and generate massive uncaptured runoff—the volume is distributed evenly across the days within the event interval. Crucially, this traditional heuristic is spatially uniform; it is entirely unaware of the topographical cascade routing. The calculated daily rate is dispatched identically to all agents n, strictly bounded by the physical actuator cap $u_{\max}$:

$$u_k^n = \min\left(\frac{w_j \cdot W_{\text{total}}}{\text{interval}},\ u_{\max}\right) \quad \forall n \in \{1, \ldots, N\} \tag{9}$$

Furthermore, a strict accounting mechanism ensures the cumulative applied irrigation does not exceed W_{total} in instances where a truncated season or floating-point drift alters the total sum. By comparing the spatially aware MPC and SAC against this spatially blind C_2 baseline, the exact agronomic value of topographical awareness can be isolated and quantified.

4 Model Predictive Control (MPC) Architecture

4.1 The Multi-Objective Cost Function

The MPC optimization problem is formulated to maximize crop yield while minimizing resource expenditure and environmental stress over a prediction horizon H_p. To ensure stable convergence within the interior-point solver (IPOPT), the cost function J is heavily normalized. Each term is scaled to $\mathcal{O}(1)$ using agronomically and economically derived reference values. The total

objective function is defined as a weighted sum of five normalized terms:

$$J = J_{\text{biomass}} + J_{\text{water}} + J_{\text{drought}} + J_{\text{ponding}} + J_{\Delta u} \tag{10}$$

4.1.1 Terminal Biomass (Mayer Cost)

To incentivize crop growth without requiring the solver to look ahead to the end of the 93-day season, the controller maximizes the field-mean biomass x_4 at the end of the prediction horizon. This is formulated as a negative cost to convert the maximization into a minimization compatible with the NLP solver:

$$J_{\text{biomass}} = -\alpha_1 \frac{\bar{x}_4(k + H_p)}{x_{4,\text{ref}}} \tag{11}$$

where $\bar{x}_4(k + H_p) = \frac{1}{N}\sum_{n=1}^{N} x_4^n(k + H_p)$ is the field-mean biomass at the terminal step of the prediction horizon, $x_{4,\text{ref}} = 900$ g/m^2 represents the target yield threshold for Hashemi rice under optimal conditions (corresponding to approximately 3,780 kg/ha via $Y = x_4 \times \text{HI} \times 10 = 900 \times 0.42 \times 10$), and $\alpha_1 = 1.0$ serves as the primary economic anchor representing the high market revenue of the rice crop.

4.1.2 Water Consumption Penalty

The cost of irrigation is penalized over the prediction horizon to enforce water conservation:

$$J_{\text{water}} = \alpha_2 \sum_{j=0}^{H_p-1} \frac{\sum_{n=1}^{N} u^n(k + j)}{W_{\text{daily,ref}}} \tag{12}$$

where $W_{\text{daily,ref}} = 5.0 \cdot N = 650$ mm represents the theoretical baseline daily demand for the $N = 130$ agents (based on the mean seasonal $\text{ET}_c \approx 5.8$ mm/day, rounded to 5.0 mm for normalization). The nominal weight $\alpha_2 = 0.01$ is anchored to the domestic water pricing structure in Iran.

4.1.3 Drought Stress Regularization

While the biological model penalizes growth during dry periods through the drought stress function h_3, an explicit penalty is added to the controller objective to aggressively prevent soil moisture (x_1) from dropping below the stress threshold (ST):

$$J_{\text{drought}} = \alpha_3 \sum_{j=0}^{H_p-1} \frac{1}{N} \sum_{n=1}^{N} \frac{\max(0, ST - x_1^n(k + j))}{ST - WP} \tag{13}$$

where $ST = (\theta_6 - p \cdot (\theta_6 - \theta_2)) \cdot \theta_5 = (0.35 - 0.20 \times 0.20) \times 400 = 124$ mm is the management allowable depletion threshold for rice ($p = 0.20$ per FAO-56), and $WP = \theta_2 \cdot \theta_5 = 0.15 \times 400 =$

60 mm is the permanent wilting point. The weight $\alpha_3 = 0.1$ ensures the controller proactively irrigates before severe deficit occurs, without dominating the biomass incentive.

4.1.4 Ponding Penalty

Due to the topographical cascade routing, excessive rainfall and over-irrigation can cause temporary localized flooding, particularly at low-elevation agents. Following the fractional routing fix (terrain.py v2.0), which eliminated true sink agents by allowing off-farm drainage through the padded DEM, the ponding penalty is applied to the field-mean surface ponding across all 130 agents:

$$J_{\text{ponding}} = \alpha_4 \sum_{j=0}^{H_p-1} \frac{1}{N} \sum_{n=1}^{N} \frac{x_5^n(k+j)}{x_{5,\text{ref}}} \tag{14}$$

where $x_{5,\text{ref}} = 50.0$ mm serves as the normalization reference representing acute storm ponding depth, and $\alpha_4 = 0.5$ places a high priority on preventing persistent waterlogging across the field. The choice of $x_{5,\text{ref}} = 50$ mm (increased from an initial value of 10 mm used in an earlier sink-only formulation) reflects the realistic ponding depths observed during monsoon-season storm events in Gilan, ensuring that the penalty term remains $\mathcal{O}(1)$ during typical wet-weather conditions rather than saturating the cost function.

4.1.5 Actuator Smoothing (Control-Rate Regularization)

To prevent erratic, high-frequency oscillatory irrigation commands across consecutive days—which would be physically damaging to valve actuators and operationally impractical—a standard control-rate regularization term is included:

$$J_{\Delta u} = \alpha_5 \sum_{j=1}^{H_p-1} \frac{\|\mathbf{u}(k+j) - \mathbf{u}(k+j-1)\|^2}{u_{\max}^2 \cdot N} \tag{15}$$

where the normalization by $u_{\max}^2 \cdot N$ ensures this term remains $\mathcal{O}(1)$ even at maximum actuator swing. The weight $\alpha_5 = 0.005$ provides gentle smoothing without materially constraining the optimizer's freedom to respond to sudden rainfall events.

4.2 Weight Calibration and Economic Anchoring

The five cost weights $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5\}$ are not arbitrary tuning parameters but are calibrated to reflect real economic and agronomic trade-offs. The primary anchor is $\alpha_1 = 1.0$, representing the revenue from a hectare of Hashemi rice at typical Gilan market prices (∼4,000 USD/ha). The water cost weight α_2 is set relative to this anchor based on the volumetric price of irrigation water in Iran. To evaluate the sensitivity of the MPC policy to economic assumptions about water pricing, three values of α_2 are tested:

Table 2: Water cost weight sweep for sensitivity analysis.

α_2 **Value**	**Water Price Tier**	**Economic Interpretation**
0.0001	Subsidized	Near-zero marginal cost of water; mimics the current heavily subsidized pumping regime in Iran
0.01	Nominal	Reflects the domestic water pricing tier; the default operating point
0.03	Scarcity premium	Reflects a future policy scenario with volumetric water pricing at cost-recovery levels

The complete weight vector at the nominal operating point is $\boldsymbol{\alpha} = \{1.0, 0.01, 0.1, 0.5, 0.005\}$. The sensitivity sweep is conducted on the dry/100% scenario at $H_p = 8$ across 7 configurations (varying α_2 while holding other weights fixed) to quantify the Pareto frontier between yield and water consumption.

4.3 Control-Oriented Model (COM) and Symbolic Formulation

To enable real-time numerical optimization within the receding horizon framework, the high-fidelity Agent-Based Model (ABM) described in the preceding chapter must be translated into a Control-Oriented Model (COM). The COM is implemented as a CasADi symbolic computational graph using the SX (Scalar eXpression) framework, defining the explicit mapping $f_{\text{COM}} : (\mathcal{X}_k, \mathcal{U}_k, \mathcal{P}_k) \to \mathcal{X}_{k+1}$, where $\mathcal{P}_k$ represents exogenous parameters (weather forecasts and precomputed biological quantities). To ensure tractability for the IPOPT interior-point solver, several rigorous dimensionality reduction and computational strategies are applied.

4.3.1 Dimensionality Reduction of the State Space

While the complete biological system tracks five state variables per agent, maintaining all variables as shooting states in the Non-Linear Programming (NLP) formulation would result in $5 \times 130 \times H_p = 5{,}200$ shooting variables at $H_p = 8$ alone, producing prohibitively large and dense Jacobian matrices. Because thermal time (x_2) is purely driven by exogenous climate data and is therefore entirely decoupled from the control action, it is precomputed offline for the full season and injected as a time-varying parameter. Maturity (x_3) acts as a slow-moving, monotonic biological clock that is tracked from the plant's true state at each receding-horizon step to prevent open-loop drift. Biomass (x_4) is accumulated inline within the COM strictly to evaluate the terminal Mayer cost but is not used as a shooting state. The COM therefore restricts the formal shooting states exclusively to the fast-moving hydrological variables:

$$\mathcal{X}_{\text{shoot}} = \{x_1, x_5\} \tag{16}$$

This reduces the shooting variables from $5N$ to $2N = 260$ per horizon step, yielding a total of

$2N \times H_p + N \times H_p = 3N \times H_p$ NLP variables (two states plus one control per agent per step) plus the initial state parameters.

4.3.2 Deterministic Caching of Biological Nonlinearities

The biological penalty functions for heat stress (h_2), cold stress (h_7), thermal time accumulation (h_1), and the base growth function (g_{base}) rely on highly nonlinear sigmoid formulations. Because these functions depend strictly on deterministic meteorological variables (temperature and radiation) and are entirely decoupled from the control action (u), they are precomputed offline for the entire 93-day season. By injecting these precomputed scalar arrays into the CasADi graph as time-varying parameters rather than calculating them inline, the NLP solver bypasses the expensive evaluation of sigmoidal gradients and their second derivatives, significantly accelerating the per-step solve time and improving Hessian sparsity.

4.3.3 Static Embedding of Topographical Cascade Routing

A primary challenge in agricultural MPC is simulating spatial water routing without relying on computationally prohibitive coupled Partial Differential Equations (PDEs). The COM resolves this by statically embedding the field's directed topographical graph into the symbolic formulation. At NLP construction time, the 130 agents are sorted into a strict topological order (from maximum elevation to terminal boundary agents). During the symbolic forward pass of each horizon step, surface hydrology is unrolled sequentially: for a given agent n processed at position p in the topological order, all uphill contributions from agents at positions $1, \ldots, p-1$ have already been computed and are available as symbolic expressions.

The surface water availability for agent n is computed as:

$$W_{\text{surf}}^{(n)}(k) = x_5^{(n)}(k) + P(k) + u^{(n)}(k) + \sum_{m \in \mathcal{U}(n)} \frac{\phi_2^{(m)}(k)}{N_r^{(m)}} \tag{17}$$

where $\mathcal{U}(n)$ is the set of uphill neighbors of agent n, and $N_r^{(m)}$ is the *total* number of lower-elevation neighbors of agent m, computed using the padded DEM (terrain.py v2.0). This fractional routing boundary condition is critical: the DEM is padded by one cell on each edge by extrapolating the slope. N_r counts all lower neighbors including external (off-farm) pad cells, while the routing loop only sends water to internal agents. When the COM divides ϕ_2 by N_r, the fraction directed toward pad cells is implicitly removed from the mass balance, accurately simulating off-farm drainage without introducing dummy variables to the solver. This fixed the "bathtub effect" observed in the initial implementation, where three sink agents ($N_r = 0$) trapped all cascade runoff and experienced catastrophic waterlogging.

4.3.4 Smooth Approximations for Solver Convergence

The original ABM equations contain several non-smooth operations—$\max(0, x)$, $\min(a, b)$, and conditional branching via `ca.if_else`—that create discontinuities in the first and second derivatives of the NLP objective and constraints. While CasADi can represent these operations

symbolically, the IPOPT interior-point solver requires C^2-continuous (twice continuously differentiable) functions to reliably compute the Hessian of the Lagrangian and perform Newton steps. Using exact non-smooth operators caused severe convergence failures: IPOPT would stall at "Restoration Failed" or "Search Direction Becomes Too Small" after encountering kinks in the derivative landscape.

To resolve this, smooth C^2-continuous approximations are employed with a smoothing parameter $\varepsilon = 0.01$:

Smooth maximum: The half-rectifier $\max(0, x)$ is replaced by:

$$\widetilde{\max}(0, x) = \frac{x + \sqrt{x^2 + \varepsilon^2}}{2} \tag{18}$$

This function satisfies $\widetilde{\max}(0, x) \to \max(0, x)$ as $\varepsilon \to 0$, is everywhere differentiable, and introduces an error bounded by $\varepsilon/2 = 0.005$ mm—negligible relative to the soil moisture range of [60, 200] mm.

Smooth minimum: The $\min(a, b)$ operation is derived from the smooth maximum via:

$$\widetilde{\min}(a, b) = a + b - \widetilde{\max}(a, b) = \frac{a + b - \sqrt{(a - b)^2 + \varepsilon^2}}{2} \tag{19}$$

Guarded denominator for drought stress: The drought stress function $h_3 = 1 - \theta_{14} \cdot \max(1 - \phi_1/\mathrm{ET}_c, 0)$ contains a division by ET_c that approaches zero on cool, humid days when the atmospheric evaporative demand vanishes. To prevent division by zero in the symbolic graph, the denominator is guarded:

$$h_3^{(n)}(k) = 1 - \theta_{14} \cdot \widetilde{\max}\left(1 - \frac{\phi_1^{(n)}(k)}{\mathrm{ET}_c(k) + \varepsilon},\ 0\right) \tag{20}$$

SCS runoff smoothing: The conditional SCS curve number computation (Eq. 5 in the ABM chapter) originally used `ca.if_else`, which creates a derivative discontinuity at $W_{\mathrm{surf}} = \theta_3$. This is replaced by the smooth formulation:

$$\phi_2^{(n)} = \frac{\widetilde{\max}(0, W_{\mathrm{surf}}^{(n)} - \theta_3)^2}{W_{\mathrm{surf}}^{(n)} + 4\theta_3 + \varepsilon} \tag{21}$$

The value $\varepsilon = 0.01$ was selected through systematic experimentation: values below 0.001 reintroduced convergence failures due to near-discontinuous derivatives, while values above 0.1 introduced unacceptable mass-balance artifacts (artificial water creation exceeding 0.5 mm/season). At $\varepsilon = 0.01$, the total mass-balance error is bounded below 0.05 mm over the 93-day season, which is four orders of magnitude smaller than the typical seasonal water budget.

4.4 NLP Construction and Solver Configuration

4.4.1 Decision Variable and Constraint Structure

The NLP is constructed using a direct multiple-shooting formulation. At each prediction horizon step $j \in \{0, \ldots, H_p - 1\}$, the decision variables consist of the shooting states $\{x_1^n(k+j), x_5^n(k+j)\}$ for all $N = 130$ agents and the control inputs $\{u^n(k+j)\}$ for all agents. The complete variable vector $\mathbf{z}$ is organized as:

$$\mathbf{z} = \left[\underbrace{x_1^{1:N}(k), x_5^{1:N}(k)}_{\text{initial states (fixed)}}, \underbrace{u^{1:N}(k), x_1^{1:N}(k+1), x_5^{1:N}(k+1), u^{1:N}(k+1), \ldots}_{\text{free variables}} \right] \tag{22}$$

For prediction horizon $H_p = 8$, this yields the following NLP dimensions:

Table 3: NLP dimensions for $H_p = 8$ and $H_p = 14$.

Quantity	$H_p = 8$	$H_p = 14$
Total symbolic variables	3,380	5,720
Free decision variables	3,120	5,460
Equality constraints (shooting gaps)	2,080	3,640
Inequality constraint (budget)	1	1
Total constraints	2,081	3,641
Box bounds on u	$2 \times 130 \times H_p$	$2 \times 130 \times H_p$
NLP build time (typical)	20–50 s	37–44 s

The 260 initial-state variables ($x_1^{1:N}(k)$ and $x_5^{1:N}(k)$) are fixed to the current true plant state at each receding-horizon step (they appear as parameters, not free variables). The 2,080 equality constraints at $H_p = 8$ enforce the shooting gap closure: each constraint requires that the symbolic COM-predicted next state matches the shooting variable at the subsequent step, i.e., $x_s^n(k+j+1) = f_{\text{COM},s}^n(k+j)$ for $s \in \{1, 5\}$, $n \in \{1, \ldots, N\}$, and $j \in \{0, \ldots, H_p - 2\}$. The single inequality constraint enforces the remaining seasonal water budget.

4.4.2 IPOPT Solver Configuration and Termination Criteria

The NLP is solved using IPOPT (Interior Point OPTimizer) version 3.14, an open-source primal-dual interior-point algorithm for large-scale nonlinear optimization. IPOPT implements a line-search filter method based on the barrier reformulation of the KKT (Karush–Kuhn–Tucker) optimality conditions. At each iteration, IPOPT solves a symmetric indefinite linear system to obtain the Newton search direction, using the MUMPS (MUltifrontal Massively Parallel Sparse direct Solver) as the default linear algebra backend.

The solver is configured with the following key parameters:

Table 4: IPOPT solver configuration.

Option	Value	Description
`tol`	10^{-8}	Desired convergence tolerance (relative). The algorithm terminates successfully when the scaled NLP error falls below this threshold.
`acceptable_tol`	10^{-6}	Acceptable convergence tolerance. If the algorithm encounters `acceptable_iter` consecutive iterates satisfying this relaxed tolerance, it terminates.
`acceptable_iter`	15	Number of consecutive acceptable iterates required to trigger early termination at the acceptable tolerance level.
`max_iter`	3,000	Maximum number of interior-point iterations per NLP solve.
`constr_viol_tol`	10^{-4}	Maximum allowable constraint violation (absolute) for successful termination.
`dual_inf_tol`	1.0	Maximum allowable dual infeasibility (absolute).
`compl_inf_tol`	10^{-4}	Maximum allowable complementarity violation.
`print_level`	0	Suppress IPOPT console output during batch runs.
`linear_solver`	MUMPS	Sparse direct solver for the KKT system.

4.4.3 IPOPT Termination Status Codes

Understanding the termination status is critical for interpreting MPC results. IPOPT reports one of several exit conditions, of which two are relevant to the completed simulations:

Solve_Succeeded (Return code 0): This status indicates that IPOPT found a locally optimal point satisfying *all* desired convergence criteria simultaneously. Specifically, the following conditions are met: (i) the scaled NLP error (a composite measure of optimality, feasibility, and complementarity, denoted E_μ in the IPOPT formulation) is smaller than `tol` $= 10^{-8}$; (ii) the max-norm of the unscaled dual infeasibility is below `dual_inf_tol` $= 1.0$; (iii) the max-norm of the unscaled constraint violation is below `constr_viol_tol` $= 10^{-4}$; and (iv) the max-norm of the unscaled complementarity is below `compl_inf_tol` $= 10^{-4}$. When all four conditions hold, the returned solution represents a KKT point of the NLP to the desired numerical precision.

Solved_To_Acceptable_Level (Return code 1): This status indicates that the algorithm

could not converge to the "desired" tolerances specified by `tol`, but it did find a point satisfying the relaxed "acceptable" tolerance level. Specifically, the IPOPT iteration encountered `acceptable_iter` = 15 consecutive iterations during which the scaled NLP error remained below `acceptable_tol` = 10^{-6} (two orders of magnitude more relaxed than the desired 10^{-8}), the constraint violation remained below `acceptable_constr_viol_tol` = 0.01 (two orders of magnitude more relaxed than the desired 10^{-4}), and the complementarity remained below `acceptable_compl_inf_tol` = 0.01. This outcome typically occurs when the NLP is poorly conditioned at a particular time step—for example, when a sudden large rainfall event creates a near-discontinuity in the optimal irrigation profile despite the smooth approximations.

The practical impact of `Solved_To_Acceptable_Level` on irrigation quality is minimal in this application. The constraint violation tolerance is relaxed from 10^{-4} mm to 10^{-2} mm, which for a soil moisture range of [60, 200] mm represents less than 0.01% relative error. In the completed simulation grid, this status was observed on only 3 out of the 90 daily solves across 9 scenario×budget combinations at $H_p = 8$: specifically at day 30 of wet/100%, day 30 of moderate/100%, and day 60 of moderate/70%. Inspection of the corresponding u_{mean} values shows no anomalous behavior, confirming that the acceptable-level solutions are operationally indistinguishable from fully converged solutions.

Other possible status codes (not observed in the completed simulations but handled by the fail-safe logic) include: `Maximum_Iterations_Exceeded` (code −1), `Restoration_Failed` (code −2), `Infeasible_Problem_Detected` (code 2), `Search_Direction_Becomes_Too_Small` (code 3), and `Error_In_Step_Computation` (code −3). In any of these failure modes, the controller defaults to a zero-irrigation fail-safe ($u_k^n = 0$ for all n), allowing the physical crop to survive on natural rainfall until the receding horizon re-establishes a feasible trajectory on the subsequent day.

4.5 Prediction Horizon Selection

The prediction horizon H_p governs the trade-off between the MPC's look-ahead depth and its computational cost. Two values are evaluated in this research:

$H_p = 8$ **days:** This horizon spans approximately one week of future weather, corresponding to the practical accuracy limit of operational meteorological forecasts. At this horizon, the NLP contains 3,120 free decision variables and 2,081 constraints, with typical build times of 20–50 seconds and per-step solve times ranging from 1.5 s to 508 s (mean: 10–54 s depending on scenario and budget).

$H_p = 14$ **days:** This extended horizon doubles the look-ahead, allowing the MPC to plan across a full two-week window. The NLP grows to 5,460 free variables and 3,641 constraints. Per-step solve times increase substantially, with means of 45–50 s and worst-case solves exceeding 360 s. The critical question is whether this additional computational investment translates into improved agronomic outcomes.

The preliminary results at 70% budget indicate that extending the horizon from 8 to 14 days provides *no measurable yield improvement* (e.g., dry/70%: 3,779 kg/ha at $H_p = 8$ versus

3,766 kg/ha at $H_p = 14$), while total runtime increases by 4–5× (e.g., dry/70%: 957 s at $H_p = 8$ versus 4,251 s at $H_p = 14$). Under severe budget constraints, the optimizer exhausts the water budget by approximately day 65, after which the remaining solve steps are trivially $u = 0$ regardless of horizon length. This finding has important practical implications: for real-time edge deployment scenarios, the shorter $H_p = 8$ horizon is strongly preferred, as it achieves equivalent agronomic performance at a fraction of the computational cost.

4.6 Warm-Starting and Computational Acceleration

To reduce solve times across consecutive daily solves, the NLP is warm-started by shifting the previous day's optimal trajectory forward by one step. The control sequence $\{u^*(k), u^*(k+1), \ldots, u^*(k+H_p-1)\}$ from the previous solve is shifted to provide an initial guess $\{u^*(k+1), u^*(k+2), \ldots, u^*(k+H_p-1), u^*(k+H_p-1)\}$ for the current solve, where the final element is duplicated. The corresponding state trajectory is similarly shifted. This provides IPOPT with a feasible starting point close to the optimal solution, typically reducing the iteration count from 200+ (cold start) to 30–80 (warm start).

5 Soft Actor-Critic (SAC) Architecture

While Model Predictive Control provides a mathematically rigorous, constraint-aware optimization framework, the requirement to solve a massive Non-Linear Program at each time step poses significant latency challenges for low-power edge computing devices deployed in agricultural Cyber-Physical Systems. To investigate a low-latency alternative, a Reinforcement Learning agent is trained to approximate the optimal control policy. The problem is formulated as a Markov Decision Process (MDP) and solved using the Soft Actor-Critic (SAC) algorithm, an off-policy actor-critic method based on the maximum entropy reinforcement learning framework.

5.1 Markov Decision Process (MDP) Formulation

The irrigation control problem is defined by the MDP tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$.

5.1.1 Observation Space ($\mathcal{S}$)

To provide the RL agent with sufficient information to manage both spatial topographical heterogeneity and the global seasonal water budget, the state observation is defined as a flat, continuous vector of 660 dimensions. The vector concatenates 10 global field features with 5 local features for each of the $N = 130$ agents:

Global Features (10 dimensions): Normalized remaining budget ($W_{\text{remaining}}/W_{\text{total}}$), normalized current day (k/K), daily rainfall $P(k)$, reference evapotranspiration $\text{ET}_0(k)$, field-mean soil moisture ($\bar{x}_1/FC$), field-moisture standard deviation, field-mean biomass ($\bar{x}_4/x_{4,\text{ref}}$), fraction of agents in drought ($x_1 < ST$), fraction of agents in waterlogging ($x_1 > FC$), and the cumulative 7-day rainfall forecast ($\sum_{j=1}^{7} P(k+j)$).

Local Features (650 dimensions): For each agent n, the network observes its normalized soil moisture (x_1^n/FC), normalized thermal time (x_2^n/θ_{18}), normalized biomass ($x_4^n/x_{4,\text{ref}}$), normalized surface ponding ($x_5^n/x_{5,\text{ref}}$), and its static normalized topographical elevation (γ^n). The elevation feature enables the policy to learn spatially differentiated irrigation strategies—for example, reducing irrigation for high-elevation agents whose runoff will cascade to lower neighbors.

5.1.2 Action Space ($\mathcal{A}$)

The continuous action space mirrors the MPC actuator constraints. The SAC actor outputs a bounded vector $\mathbf{a}_k \in [0,1]^N$ via a tanh squashing function. During the environment step, this output is linearly scaled by the physical actuator cap ($u_{\max} = 12.0$ mm/day) such that $u_k^n = a_k^n \cdot u_{\max}$, matching the exact decision space of the MPC baseline.

5.1.3 Reward Formulation ($\mathcal{R}$)

To ensure a fair and direct comparison between the learned policy and the computed MPC policy, the dense per-step reward r_k is engineered as the exact mathematical negation of the MPC path cost function (Section 4.1):

$$r_k = -\left(J_{\text{water}}(k) + J_{\text{drought}}(k) + J_{\text{ponding}}(k)\right) \tag{23}$$

At the terminal time step $K = 93$, a sparse positive bonus is awarded based on the normalized terminal biomass:

$$r_K = +5 \cdot \frac{\bar{x}_4(K)}{x_{4,\text{ref}}} \tag{24}$$

The terminal bonus weight of +5 was selected to ensure that the cumulative terminal reward is of the same order of magnitude as the sum of dense path rewards over 93 steps, preventing the agent from myopically minimizing path costs at the expense of long-term biomass accumulation. This reward structure forces the agent to learn the exact same agronomical and economic trade-offs as the predictive optimizer.

5.2 Gymnasium Environment Wrapper

The ABM is wrapped as a standard Gymnasium (formerly OpenAI Gym) environment implementing the `reset()`, `step()`, and `render()` interface. The `reset()` method initializes the 130-agent field to the standard initial conditions ($x_1 = 140$ mm, $x_5 = 0$, $x_2 = 210$, $x_4 = 60$, $x_3 = 0$), samples a climate scenario, and resets the budget counter. The `step(action)` method: (i) clips the raw action to satisfy the actuator bounds and remaining budget, (ii) applies the irrigation to the ABM, (iii) advances the ABM by one day including cascade routing, (iv) constructs the 660-dimensional observation vector, (v) computes the dense reward, and (vi) returns the standard Gymnasium tuple $(\text{obs}, \text{reward}, \text{terminated}, \text{truncated}, \text{info})$. The episode terminates naturally at $k = 93$ or is truncated early if the budget violation penalty is triggered.

5.3 Centralized Training with Decentralized Execution (CTDE)

A naive multi-agent implementation—where a single monolithic policy network directly outputs 130 independent actions from the 660-dimensional state—would fail to exploit the spatial symmetry of the field and require an intractable number of parameters. To overcome this, the architecture employs Centralized Training with Decentralized Execution (CTDE):

The Decentralized Actor: The actor network operates on the principle of parameter sharing. A single Multi-Layer Perceptron (MLP) with hidden layers of [256, 256] and ReLU activations is instantiated. During the forward pass, this network is applied identically across all 130 agents. For agent n, the actor concatenates the 10 global features with the 5 local features of agent n (totaling 15 inputs) to output a parameterized Gaussian distribution (mean μ^n and log standard deviation $\log \sigma^n$) for that specific agent's irrigation action. This forces the network to learn a generalized policy: *how to irrigate any agent given its specific elevation and moisture state, conditioned on the global budget status.*

The Centralized Critic: Unlike the actor, the twin critic networks evaluate the joint action value $Q(\mathbf{S}_k, \mathbf{A}_k)$. Each critic takes the full 660-dimensional state and the 130-dimensional joint action vector as input (790 dimensions total), passing through hidden layers of [512, 512, 256] with ReLU activations. By observing the entire state-action space during offline training, the centralized critic learns the shadow price of the global water budget and the cascading runoff effects across the topographical graph, guiding the decentralized actor toward a globally optimal cooperative policy. The twin-critic architecture (two independent Q-networks with minimum taken for the target) follows the standard SAC formulation to mitigate overestimation bias.

5.4 Budget Constraint Handling

While IPOPT natively handles the global water budget $\sum u \leq W_{\text{total}}$ via a hard linear inequality constraint in the NLP, standard RL algorithms lack native constrained-optimization mechanisms. The SAC environment enforces the budget via a three-tier soft penalty system:

Tier 1 — Action Clipping (Hard Safety Net): As an absolute guarantee, the environment clips the daily action array so that the total requested volume across all agents never mathematically exceeds the remaining budget. Specifically, if $\sum_n u_k^n > W_{\text{remaining}}(k)$, all actions are proportionally scaled down: $u_k^n \leftarrow u_k^n \cdot W_{\text{remaining}}(k) / \sum_n u_k^n$. This ensures that the budget constraint is never violated, regardless of the agent's policy output.

Tier 2 — Burn-Rate Shaping: To prevent the agent from dumping its entire budget in the first few weeks (a common failure mode during early training), a negative reward shaped by the normalized season progression is applied if the daily expenditure rate significantly outpaces the seasonal timeline. Let $f_{\text{spent}} = 1 - W_{\text{remaining}}/W_{\text{total}}$ be the fraction of budget spent and $f_{\text{time}} = k/K$ be the fraction of the season elapsed. If $f_{\text{spent}} > f_{\text{time}} + 0.15$ (the agent is spending more than 15 percentage points ahead of schedule), a penalty of $r_{\text{burn}} = -2.0 \cdot (f_{\text{spent}} - f_{\text{time}})$ is applied.

Tier 3 — Early Termination: If the agent triggers the hard clip limit by attempting to exceed the total budget, a severe penalty ($r = -100$) is applied, and the episode is immediately

truncated. This catastrophic penalty signal provides a strong gradient during early training to discourage budget-violating policies.

5.5 Training Protocol and Hyperparameters

The SAC agent is trained using Stable-Baselines3 (SB3), a well-maintained PyTorch implementation of standard RL algorithms. Training is conducted on Kaggle's free GPU tier, which provides NVIDIA Tesla P100 GPUs with a 9-hour session limit and 30 GPU-hours per week. The complete hyperparameter configuration is:

Table 5: SAC training hyperparameters.

Hyperparameter	**Value**	**Source/Justification**
Learning rate (actor & critic)	3×10^{-4}	Default SAC
Replay buffer size	10^6 transitions	Standard for continuous control
Batch size	256	Standard for continuous control
Discount factor (γ)	0.99	Near-horizon: 93-step episodes
Soft update coefficient (τ)	0.005	Default SAC
Entropy coefficient (α)	Auto-tuned	Constrained dual gradient descent
Target entropy	$-\dim(\mathcal{A}) = -130$	Default: $-\lvert\mathcal{A}\rvert$
Total training steps	5×10^6	Kaggle budget permitting
Random seeds	5	Statistical significance

The entropy coefficient α is automatically tuned via the constrained formulation introduced by Haarnoja et al. (2019), where α is treated as a Lagrange multiplier on the minimum entropy constraint $\mathcal{H}[\pi] \geq \mathcal{H}_{\text{target}}$. The target entropy is set to -130 (one per action dimension), following the standard heuristic of $-\dim(\mathcal{A})$.

Training is conducted for 5 independent random seeds on the dry/100% scenario. Each seed runs for up to 5×10^6 environment steps (approximately 53,763 full episodes of 93 steps each). Once trained, the best-performing seed (selected by validation reward) is evaluated across all 9 scenario$\times$budget combinations to assess generalization.

6 Uncertainty Modeling and Unified Evaluation Framework

To guarantee the validity of the comparative analysis, the experimental framework must ensure that all controllers are subjected to identical environmental conditions, constraints, and standardized evaluation metrics. Furthermore, to evaluate the real-world robustness of the predictive models (MPC and SAC), a formal mathematical model for meteorological forecast uncertainty must be established.

6.1 Meteorological Forecast Uncertainty

While the baseline controllers (C_1, C_2) and the "No-Forecast" SAC (C_5) operate without future weather information, the predictive controllers—the Perfect MPC (C_3), Noisy MPC (C_4), and Forecast SAC (C_6)—rely on a prediction horizon. To benchmark the theoretical upper bound of the predictive architecture against a realistic deployment scenario, two distinct forecast modes are formalized.

1. Perfect Forecast Mode. In the perfect information scenario, the forecast vector provided to the controller exactly matches the true future weather realization generated by the environment simulation. For any meteorological variable w at a future day $k + j$ predicted from the current day k:

$$\hat{w}(k + j|k) = w(k + j) \quad \forall j \in \{0, \ldots, H_p - 1\} \tag{25}$$

This mode establishes the absolute performance ceiling for the predictive controllers.

2. Noisy Forecast Mode. To simulate the decaying accuracy of real-world meteorological forecasts over time, a stochastically degrading noise model is applied to the rainfall and reference evapotranspiration (ET_0) vectors. Temperature forecasts are left unperturbed, reflecting the high operational accuracy of thermal forecasting (RMSE $< 1°$C at 7-day lead time). The multiplicative noise injection is defined as:

$$\hat{w}(k + j|k) = w(k + j) \cdot (1 + \epsilon_j) \tag{26}$$

where the error term ϵ_j is drawn from a Gaussian distribution with zero mean and a variance that grows proportionally to the square root of the look-ahead horizon j:

$$\epsilon_j \sim \mathcal{N}(0, \sigma_j^2), \quad \sigma_j = 0.15 \cdot \sqrt{j} \tag{27}$$

This formulation ensures that tomorrow's forecast ($j = 1$) retains high fidelity ($\sigma_1 = 0.15$, corresponding to $\pm 15\%$ typical error), while forecasts extending toward the end of a 14-day horizon exhibit severe uncertainty ($\sigma_{14} = 0.56$, corresponding to $\pm 56\%$ typical error), rigorously testing the stability of the receding-horizon control loop. The square-root decay rate is consistent with the empirical skill degradation observed in operational NWP (Numerical Weather Prediction) models.

6.2 Controller Summary Table

To provide a complete reference for the experimental evaluation, all six controllers are formally defined:

Table 6: Complete controller taxonomy.

ID	Controller	Type	Forecast	Closed-Loop
C_1	No-Irrigation Baseline	Open-loop	None	No
C_2	Fixed-Schedule Heuristic	Open-loop	None	No
C_3	MPC (Perfect Forecast)	Model-based	Perfect	Yes
C_4	MPC (Noisy Forecast)	Model-based	Noisy	Yes
C_5	SAC (No Forecast)	Learning-based	None	Yes
C_6	SAC (7-day Forecast)	Learning-based	7-day sum	Yes

6.3 Unified Software Evaluation Architecture

To eliminate implementation bias during the comparative evaluation, the control loop is strictly abstracted into three separated software layers:

The Environment Layer maintains the true physical state of the field ($\mathbf{X}_k$), executes the topographical cascade routing using the high-fidelity ABM, and calculates true biological growth. This layer is identical for all six controllers and is never modified or approximated during evaluation.

The Controller Layer provides a standardized interface. All six controllers (C_1 through C_6) inherit from an identical abstract base class, requiring them to implement a single method: `get_action(state, forecast)` $\rightarrow$ `action_vector`. This ensures that the heavy IPOPT NLP solver and the trivial zero-action baseline interact with the environment through the exact same API.

The Runner Layer is a centralized simulation orchestrator that manages the day-by-day transition loop. At each step k, it: (i) queries the instantiated controller for an action $\mathbf{U}_k$, (ii) clips the action to satisfy the remaining budget, (iii) applies the action to the environment, (iv) deducts the applied volume from the global seasonal budget counter, (v) advances the climate data to the next step, and (vi) logs the complete state, action, and diagnostic information to a long-format Parquet file with a JSON metadata sidecar.

This architecture guarantees that variations in final agronomic metrics (yield, water consumption, stress days) are strictly attributable to the intrinsic quality of the underlying control policy, not to implementation artifacts.

6.4 Experimental Grid

The complete experimental evaluation spans the following factorial design:

Table 7: Experimental grid for MPC evaluation.

	100% Budget	**85% Budget**	**70% Budget**
Dry scenario (2022)	✓	✓	✓
Moderate scenario (2020)	✓	✓	✓
Wet scenario (2024)	✓	✓	✓

Each cell is evaluated at $H_p = 8$ (complete for all 9 cells) and $H_p = 14$ (in progress for 70% budget). The noisy forecast and SAC evaluations will follow the same 9-cell grid. With the fixed-schedule baseline (C_2) already complete for all 9 cells across 3 repetitions, and the no-irrigation baseline (C_1) complete for 3 repetitions, the full evaluation comprises $6 \times 9 \times 2 = 108$ simulation runs (excluding repetitions and horizon variants).

7 Conclusion

This chapter has established the complete mathematical and architectural framework for the four irrigation controllers that will be comparatively evaluated in the subsequent results chapter. The principal design decisions and their theoretical implications are summarized as follows.

The constrained optimal control problem was formalized with a five-term multi-objective cost function, where each term is normalized to $\mathcal{O}(1)$ and anchored to real agronomic and economic quantities. The MPC controller was implemented as a large-scale NLP with 3,120 decision variables and 2,081 constraints (at $H_p = 8$), solved by the IPOPT interior-point method using smooth C^2-continuous approximations ($\varepsilon = 0.01$) to guarantee convergence. The solver achieves successful convergence (`Solve_Succeeded`) on 97% of daily solves, with the remaining 3% converging to acceptable tolerance levels (`Solved_To_Acceptable_Level`, relative tolerance relaxed from 10^{-8} to 10^{-6}) with negligible impact on irrigation quality. The fractional DEM-padded routing boundary condition resolves the bathtub effect and enables physically realistic off-farm drainage simulation within the symbolic graph.

The SAC reinforcement learning agent was designed with a CTDE architecture that exploits spatial symmetry through parameter sharing (15-dimensional per-agent input to a shared 256×256 actor), while the centralized twin-critic (790-dimensional input through 512×512×256 layers) captures the global budget dynamics and inter-agent runoff coupling. The budget constraint is enforced through a three-tier soft penalty system that provides hard safety (action clipping), temporal shaping (burn-rate penalty), and catastrophic deterrence (early termination).

The theoretical trade-off between these approaches is clear. MPC provides mathematical guarantees on constraint satisfaction and can exploit exact model knowledge and weather forecasts to compute globally optimal irrigation profiles, but at a computational cost of 10–54 seconds per daily decision (mean) with worst-case latencies exceeding 500 seconds. SAC, once trained, executes a simple feed-forward neural network inference in sub-millisecond time, enabling real-time deployment on edge computing hardware, but lacks native constraint guarantees and may

exhibit suboptimal behavior in out-of-distribution weather scenarios. The subsequent experimental chapter will quantify these trade-offs through systematic comparison across the 9-cell scenario×budget grid.

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